

SEC: Sr.Elite & Target(C-120, C-IPL & IPL-IC) JEE-MAIN

Date: 02-09-2023

Time: 09.30Am to 12.30Pm

RPTM -03

Max. Marks: 300

IMPORTANT INSTRUCTION:

1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black Ball Point Pen** only.
2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of 90 questions. The maximum marks are **300**.
5. There are **three** parts in the question paper 1,2,3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and subject having **two sections**.

(I) Section -I contains 20 **multiple choice** questions with only one correct option.

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

(II) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions only, if more than 5 questions attempted, First 5 attempted questions will be considered.

■ The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

To cancel any attempted question bubble on the question number box.

For example: To cancel attempted question 21. Bubble on 21 as shown below



Question Answered for Marking **Question Cancelled for Marking**

Marking scheme: +4 for correct answer, 0 if **not attempt** and -1 in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number:

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Candidate's Signature: _____

Invigilator's Signature: _____

02-09-2023_Sr.Elite & Target(C-120, C-IPL & IPL-IC) _Jee-Main_ RPTM -03_Test Syllabus

PHYSICS : Kinematics, NLM

CHEMISTRY : T-1 : Chemical Bonding (upto MOT)

T-2 : Alkynes, Benzene & Alkyl halides (Preparation & physical Properties)

MATHEMATICS : Applications of Differentiation (AOD)

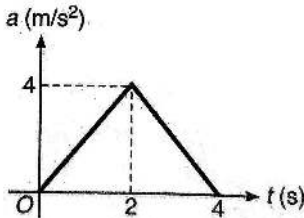
PHYSICS

Max Marks: 100

(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

- A ball is thrown upwards from the ground with an initial speed of u . The ball is at a height of 80 m at two times, the time interval being 6 s. Find u . Take $g = 10 \text{ m/s}^2$.
 1) 50 m/s 2) 20 m/s 3) 25 m/s 4) 75 m/s
- Acceleration – time graph of a particle moving in a straight line is shown in fig. Velocity of particle at time $t=0$ is 2 m/s. Find velocity at the end of fourth second.

 1) 5 m/s 2) 10 m/s 3) 15 m/s 4) 20 m/s
- The relation between time t and displacement x is $t = \alpha x^2 + \beta x$ where α and β are constants. The retardation is
 1) $2\alpha v^3$ 2) $2\beta v^3$ 3) $2\alpha\beta v^3$ 4) $2\beta^2 v^3$
- An aeroplane is flying in a horizontal direction with a velocity 600 km/h at a height of 1960 m. When it is vertically above the point A on the ground, a body is dropped from it. The body strikes the ground at point B. Calculate the distance AB.
 1) 5.55 km 2) 3.33 km 3) 4 km 4) 2.5 km
- Average velocity of a particle in projectile motion between its starting point and the highest point of its trajectory is (projection speed = u , angle of projection from horizontal = θ)
 1) $u \cos \theta$ 2) $\frac{u}{2} \sqrt{1 + 3 \cos^2 \theta}$ 3) $\frac{u}{2} \sqrt{2 + \cos^2 \theta}$ 4) $\frac{u}{2} \sqrt{1 + \cos^2 \theta}$

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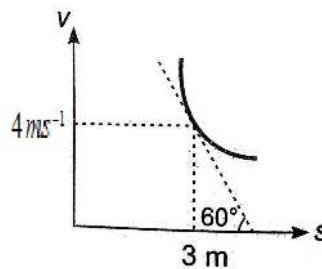
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MARKS

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6. At certain moment of time, velocities of 1 and 2 both are 1 m/s upwards. Find the velocity of 3 at that moment.



- 1) 2.5 m/s 2) 5.0 m/s 3) -3 m/s 4) -4.5 m/s
7. A particle is moving along a straight line whose velocity-displacement graph is as shown in the figure. What is the magnitude of acceleration when displacement is 3 m?



- 1) $4\sqrt{3} \text{ ms}^{-2}$ 2) $3\sqrt{3} \text{ ms}^{-2}$ 3) $\sqrt{3} \text{ ms}^{-2}$ 4) $\frac{4}{\sqrt{3}} \text{ ms}^{-2}$
8. Assertion(1): Two frames S_1 and S_2 are non-inertial. Then frame S_2 when observed from S_1 is inertial.
- Reason(R): A frame in motion is not necessarily a non-inertial frame
- 1) If both Assertion and reason are true and the reason is correct explanation of the assertion.
- 2) If both assertion and reason are true but reason is not the correct explanation of assertion
- 3) Assertion is true, reason is false.
- 4) Assertion is false, reason is true.

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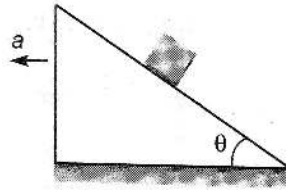
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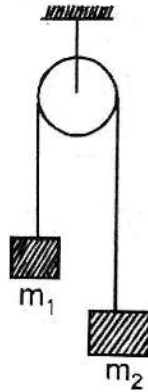
9. A block of mass m is resting on a wedge of angle θ as shown in the figure. With what minimum acceleration a should the wedge move so that the mass m falls freely ?



- 1) g 2) $g \cos \theta$ 3) $g \cot \theta$ 4) $g \tan \theta$
10. A smooth inclined plane of length L having inclination θ with the horizontal is inside a lift which is moving down with a retardation a . The time taken by a body to slide down the inclined plane from rest will be.

1) $\sqrt{\frac{2L}{(g+a)\sin\theta}}$ 2) $\sqrt{\frac{2L}{(g-a)\sin\theta}}$ 3) $\sqrt{\frac{2L}{a\sin\theta}}$ 4) $\sqrt{\frac{2L}{g\sin\theta}}$

11. Two masses $m_1 = 10 \text{ kg}$ and $m_2 = 8 \text{ kg}$ tied to a string are hanging over a light frictionless pulley. What is the acceleration of the masses when left free to move ? ($g = 9.8 \text{ m/s}^2$)



1) 5 m/s^2 2) 9.8 m/s^2 3) 1.08 m/s^2 4) 4.8 m/s^2

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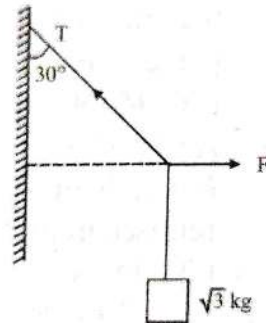
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12. A block of $\sqrt{3}kg$ is attached to a string whose other end is attached to the wall. An unknown force F is applied so that the string makes an angle of 30° with the wall. The tension T is :
(Given $g = 10 ms^{-2}$)



- 1) 20 N 2) 25 N 3) 10 N 4) 15 N
13. A particle of mass m is acted upon by a force F given by the empirical law $F = \frac{R}{t^2}v(t)$. If this law is to be tested experimentally by observing the motion starting from rest, the best way is to plot :
- 1) $\log_e v(t)$ against $\frac{1}{t}$ 2) $v(t)$ against t^2 3) $\log_e v(t)$ against $\frac{1}{t^2}$ 4) $\log_e v(t)$ against t
14. A block of mass M placed inside a box descends vertically with acceleration 'a'. The block exerts a force equal to one third of its weight on the floor of the box. The value of 'a' will be
- 1) $\frac{g}{4}$ 2) $\frac{g}{2}$ 3) $\frac{3g}{4}$ 4) $\frac{2}{3}g$
15. A balloon has mass of 10 g in air. The air escapes from the balloon at a uniform rate with velocity 2.5 cm/s. If the balloon shrinks in 5 s completely. Then, the average force acting on that balloon will be (in dyne)
- 1) 3 2) 9 3) 5 4) 18



16. An object is thrown vertically upwards. At its maximum height, which of the following quantity becomes zero ?
 1) Momentum 2) Potential energy 3) Acceleration 4) Force
17. Two projectiles are thrown with same initial velocity making an angle of 15° and 30° with the horizontal respectively. The ratio of their respective ranges will be
 1) $1:\sqrt{2}$ 2) $\sqrt{2}:1$ 3) $1:\sqrt{3}$ 4) $\sqrt{3}:1$
18. Statement I: If three force $\vec{F}_1, \vec{F}_2$ and $\vec{F}_3$ are represented by three sides of triangle and $\vec{F}_1 + \vec{F}_2 = -\vec{F}_3$, then these three forces are concurrent forces and satisfy the condition for equilibrium.
 Statement II: A triangle made up of three forces $\vec{F}_1, \vec{F}_2$ and $\vec{F}_3$ as its sides taken in the same order, satisfy the condition for translatory equilibrium.
 In the light of the above statements, choose the most appropriate answer from the options given below:
 1) Both statement I and statement II are true
 2) Statement I is true but statement II is false
 3) Both statement I and statement II are false.
 4) Statement I is false but statement II is true
19. When a ball is dropped into a lake from a height 4.9 m above the water level, it hits the water with a velocity v and then sinks to the bottom with the constant velocity v . It reaches the bottom of the lake 5.0 s after it is dropped. The approximate depth of the lake is
 1) 19.6 m 2) 29.4 m 3) 39.2 m 4) 73.5 m



20. The initial mass of a rocket is 1000 kg. Calculate at what rate the fuel should be burnt so that the rocket is given an acceleration of 20 ms^{-1} . The gases comes out at a relative speed of 500 ms^{-1} with respect to the rocket: [Use $g = 10 \text{ m/s}^2$]

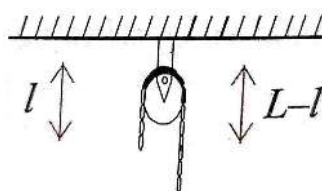
- 1) 10 kg s^{-1} 2) 60 kg s^{-1} 3) 500 kg s^{-1} 4) $6.0 \times 10^2 \text{ kg s}^{-1}$

(NUMERICAL VALUE TYPE)

Section-II contains 10 Numerical Value Type questions. Attempt any 5 questions only. First 5 attempted questions will be considered if more than 5 questions attempted. The Answer should be within 0 to 9999. If the Answer is in Decimal then round off to the nearest Integer value (Example i.e. If answer is above 10 and less than 10.5 round off is 10 and if answer is from 10.5 and less than 11 round off is 11).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

21. A uniform metal chain of mass m and length ' L ' passes over a massless and frictionless pulley. It is released from rest with a part of its length ' l ' is hanging on one side and rest of length ' $L-l$ ' is hanging on the other side of the pulley. At a certain point of time, when $l = \frac{L}{x}$, the acceleration of the chain is $\frac{g}{2}$. The value of x is _____. [Given, $L > 2l$]



22. A stone tied to 100 cm long string at its end is making 28 revolutions in horizontal circle in every minute. The magnitude of acceleration of stone is $\frac{1936}{x} \text{ ms}^{-2}$. The value of x is _____. (Take $\pi = \frac{22}{7}$)
23. A swimmer can swim with velocity of 12 km/h in still water. Water flowing in river has velocity 6 km/h. The direction with respect to the direction of flow of river water he should swim in order to reach the point on the other bank just opposite to his starting point is _____. (Round off to the Nearest Integer)

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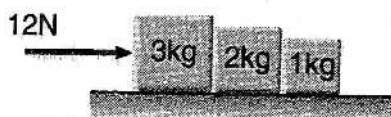
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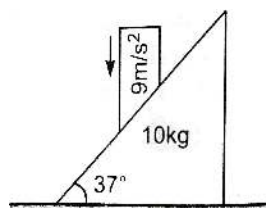


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24. The maximum height attained by projectile is increased by 10% by increasing the speed of projection without changing angle of projection. What is the percentage increase in the horizontal range attained by the projectile?
25. A batsman hits back a ball of mass 0.5 kg straight in the direction of the bowler without changing its initial speed of 16 ms^{-1} . The impulse imparted to the ball is _____Ns.
26. Three blocks of mass 3 kg, 2 kg and 1 kg are placed side by side on a smooth surface as shown in figure. A horizontal force of 12 N is applied on 3 kg block. Find the net force (in N) on 2 kg block.



27. A ball is projected with velocity 20 ms^{-1} at an angle to the horizontal. In order to have the maximum range, its velocity at the highest position must be _____m/s.
28. A ball is dropped from the top of a 100 m high tower on a planet. In the last $\frac{1}{2}s$ before hitting the ground, it covers a distance of 19 m. Acceleration due to gravity (in ms^{-2}) near the surface on that planet is _____.
29. A ball is projected upwards from the top of a tower with a velocity 50 ms^{-1} making an angle 30° with the horizontal. The height of tower is 70 m. After how many seconds from the instant of throwing, will the ball reach the ground. ($g = 10\text{ms}^{-2}$)
30. In the system shown in figure all surfaces are smooth. Rod is moved by external agent with acceleration 9 ms^{-2} vertically downwards. Force exerted on the rod by the wedge will be _____N.



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CHEMISTRY**Max Marks: 100****(SINGLE CORRECT ANSWER TYPE)**

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

31. The ion that has sp^3d hybridisation for the central atom, is
 1) $[ICl_4]^-$ 2) $[ICl_2]^-$ 3) $[IF_6]^-$ 4) $[BrF_4]^-$
32. Number of lone pair of electrons central element of iodine in I_3^- is
 1) 3 2) 6 3) 9 4) 12
33. Which of the following are isoelectronic and isostructural?
 1) ClO_3^-, SO_3 2) SO_3, NO_3^- 3) ClO_3^-, CO_3^{2-} 4) NO_3^-, CO_3^{2-}
34. The type of hybrid orbitals used by the chlorine atom in ClO_4^- is
 1) sp^3 2) sp^2 3) sp 4) None of these
35. The bond order and magnetic behavior of O_2^{2-} ion are, respectively:
 1) 1.5 and paramagnetic 2) 1.5 and diamagnetic
 3) 2 and diamagnetic 4) 1 and diamagnetic
36. Stability of the species Li_2, Li_2^- and Li_2^+ increases in the order of :
 1) $Li_2 < Li_2^+ < Li_2^-$ 2) $Li_2 < Li_2^- < Li_2^+$ 3) $Li_2^- < Li_2^+ < Li_2$ 4) $Li_2^- < Li_2 < Li_2^+$
37. The octet rule is not valid for the molecule
 1) CO_2 2) H_2O 3) O_2 4) BH_3
38. Assertion (1) : Acetylene is formed when ethylene chloride or ethylidene chloride is heated with alcoholic KOH
 Reason (R) : Both gem dihalides and vicinal dihalides on dehydrohalogenation form alkyne
 1) Both A and R are true R is the correct explanation of A
 2) Both A and R are true R is not the correct explanation of A
 3) A is true but R is false
 4) A is false but R is true

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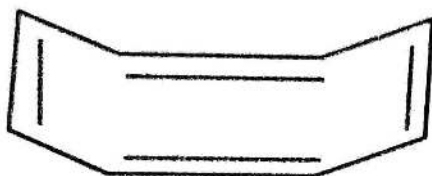
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39. Assertion (A): Reaction of but-2-yne by $Na/liqNH_3$ gives trans But-2-ene

Reason (R) : It is syn addition

- 1) Both A and R are true R is the correct explanation of A
 - 2) Both A and R are true R is not the correct explanation of A
 - 3) A is true but R is false
 - 4) A is false but R is true
40. Toluene on oxidation with dilute HNO_3 give
- 1) Benzaldehyde
 - 2) phenol
 - 3) Nitrotoluene
 - 4) Benzoic acid
41. Assertion (1): The compound cyclooctatetraene has the following structural formula :



It is cyclic with 8π -electron system but it is not an aromatic compound.

Reason (R) : $(4n+2) \pi$ electron rule does not hold good and ring is not planar.

- 1) Both A and R are true R is the correct explanation of A
 - 2) Both A and R are true R is not the correct explanation of A
 - 3) A is true but R is false
 - 4) A is false but R is true
42. Assertion (A) : Both Benzene & ethyne give same product on ozonolysis
- Reason (R) : Ethyne & benzene possess same empirical formula.
- 1) Both A and R are true R is the correct explanation of A
 - 2) Both A and R are true R is not the correct explanation of A
 - 3) A is true but R is false
 - 4) A is false but R is true

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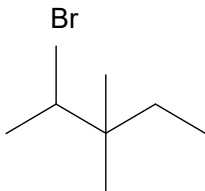
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43. In correct statement about the structure of PCl_5
- 1) axial $P-Cl$ bonds have more bond length than equatorial $P-Cl$ bonds
 - 2) d_z^2 orbital is involved in sp^3d hybridization.
 - 3) The no. of planar atoms in PCl_5 is 4.
 - 4) Phosphorous is in second excited state.
44. In the preparation of alkyl halide from alkene and halogen which of the following reaction is involved
- 1) Electrophilic addition
 - 2) Nucleophilic addition
 - 3) Electrophilic substitution
 - 4) Nucleophilic substitution
45. In the preparation of alkyl halide from alkane and halogen which of the following reaction is involved
- 1) Free radical substitution
 - 2) Nucleophilic addition
 - 3) Electrophilic substitution
 - 4) Nucleophilic substitution
46. IUPAC name of
- 
- 1) 2-bromo-3-methyl pentane
 - 2) 2-bromo-3,3-dimethyle butane
 - 3) 4-bromo-3,3-dimethyl butane
 - 4) 2-bromo-3,3-dimethyl pentane
47. Thionyl chloride is preferred in the preparation of chloro compound from alcohol since
- 1) Both the byproducts are gases and they escape out leaving product in pure state
 - 2) It is a chlorinating agent
 - 3) It is a oxidising agent
 - 4) All other reagent are unstable

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48. Correct statement about Hunsdiecker reaction is
- 1) number of carbon atoms decrease
 - 2) number of carbon atoms increase
 - 3) number of carbon atoms remains same
 - 4) carbocation intermediate is formed
49. Benzene on nitration gives nitrobenzene in presence of HNO_3 and H_2SO_4 mixture, where:
- 1) Both H_2SO_4 and HNO_3 act as a bases
 - 2) HNO_3 acts as an acid and H_2SO_4 acts as a base
 - 3) Both H_2SO_4 and HNO_3 act as an acids
 - 4) HNO_3 acts as a base and H_2SO_4 acts as an acid
50. Which of the following compounds does not dissolve in conc. H_2SO_4 even on warming?
- 1) Ethylene
 - 2) Benzene
 - 3) Hexane
 - 4) Aniline

(NUMERICAL VALUE TYPE)

Section-II contains 10 Numerical Value Type questions. Attempt any 5 questions only. First 5 attempted questions will be considered if more than 5 questions attempted. The Answer should be within 0 to 9999. If the Answer is in Decimal then round off to the nearest Integer value (Example i.e. If answer is above 10 and less than 10.5 round off is 10 and If answer is from 10.5 and less than 11 round off is 11).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

51. Based on VSEPR theory, the number of 90 degree bond angles in SF_6 is
52. According to molecular orbital theory, the number of unpaired electron(s) in O_2 is:
53. Among $H_2, He_2^+, Li_2, Be_2, B_2, C_2, N_2, O_2^-, O_2^{2-}$ and F_2 , the number of diamagnetic species is (Atomic numbers: $H = 1, He = 2, Li = 3, Be = 4, B = 5, C = 6, N = 7, O = 8, F = 9$)
54. The difference between bond orders of CO and O_2 is $\frac{x}{2}$ where $x =$ _____
55. The number of sigma bonds in allyl chloride is _____.
56. Consider the following chemical reaction. $CH \equiv CH \xrightarrow[(2) CH_3Cl, AlCl_3]{(1) Red hot Fe tube, 873K}$ product. The number of sp^2 hybridized carbon atom(s) present per molecule of the product formed is _____.

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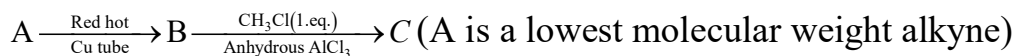
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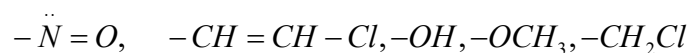
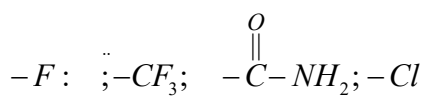
57. In the following series of reactions the maximum number of atoms present in molecule 'C' in one plane is _____.



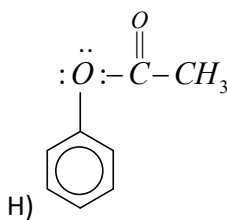
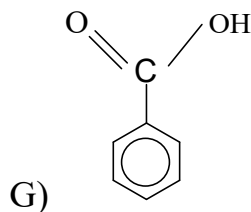
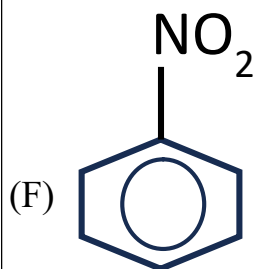
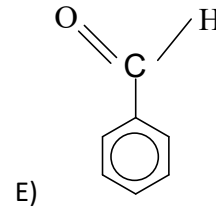
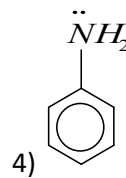
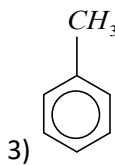
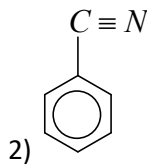
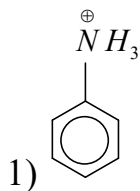
58. Among the following, the number of aromatic compound(s) is



59. Identify number of substituents which are deactivating but ortho and para directing in EAS reaction



60. Examine the structural formulas shown below and find out how many compounds cannot give Friedel crafts reaction



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MATHEMATICS

Max Marks: 100

(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

61. The radius of a right circular cylinder increases at the rate of 0.1 cm/min, and the height decreases at the rate of 0.2 cm/min. The rate of change of the volume of the cylinder, in cm^3 / min , when the radius is 2 cm and the height is 3 cm is
- 1) -2π 2) $-\frac{8\pi}{5}$ 3) $-\frac{3\pi}{5}$ 4) $\frac{2\pi}{5}$
62. If m is the minimum value of k for which the function $f(x) = x\sqrt{kx - x^2}$ is increasing in the interval $[0, 3]$ and M is the maximum value of f in $[0, 3]$ when $k = m$, then the ordered pair (m, M) is equal to
- 1) $(4, 2\sqrt{2})$ 2) $(4, 3\sqrt{3})$ 3) $(3, 3\sqrt{3})$ 4) $(5, 2\sqrt{6})$
63. If at each point of the curve $y = x^3 - ax^2 + x + 1$, the tangent is inclined at an acute angle with the positive direction of the x-axis, then
- 1) $a > 0$ 2) $a \leq \sqrt{3}$ 3) $-\sqrt{3} \leq a \leq \sqrt{3}$ 4) $a > \sqrt{3}$
64. If the absolute maximum value of the function $f(x) = (x^2 - 2x + 7)e^{(4x^3 - 12x^2 - 180x + 31)}$ in the interval $[-3, 0]$ is $f(\alpha)$, then
- 1) $\alpha = 0$ 2) $\alpha = -3$ 3) $\alpha \in (-1, 0)$ 4) $\alpha \in (-3, -1)$
65. The value of c in the Lagranges mean value theorem for the function $f(x) = x^3 - 4x^2 + 8x + 11$, when $x \in [0, 1]$ is
- 1) $\frac{2}{3}$ 2) -3 3) $\frac{4 - \sqrt{7}}{3}$ 4) $\frac{4 - \sqrt{5}}{6}$

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66. The local maximum value of the function $f(x) = \left(\frac{2}{x}\right)^{x^2}$, $x > 0$,
- 1) $2+e$ 2) $\left(\frac{4}{\sqrt{e}}\right)^{\frac{e}{4}}$ 3) $(e)^{\frac{2}{e}}$ 4) $(2\sqrt{e})^{\frac{1}{e}}$
67. The surface area of a balloon of spherical shape being inflated, increases at a constant rate. If initially, the radius of balloon is 3 units and after 5 seconds, it becomes 7 units, then its radius after 9 seconds is
- 1) 9 2) 20 3) 21 4) 12
68. Let f be a twice differentiable function on $(1,6)$. If $f(2)=8$, $f'(2)=5$, $f'(x) \geq 1$ and $f''(x) \geq 4$, for all $x \in (1,6)$, then
- 1) $f(5) \leq 10$ 2) $f'(5) + f''(5) \leq 20$
 3) $f(5) + f'(5) \geq 28$ 4) $f(5) + f'(5) \leq 26$
69. The surface area of a cube is increasing at the rate of $2 \text{ cm}^2 / \text{sec}$. When its edge is 90 cm, the volume is increasing at the rate of
- 1) $1620 \text{ cm}^3 / \text{sec}$ 2) $810 \text{ cm}^3 / \text{sec}$ 3) $405 \text{ cm}^3 / \text{sec}$ 4) $45 \text{ cm}^3 / \text{sec}$
70. The global maximum value of $f(x) = \log_{10}(4x^3 - 12x^2 + 11x - 3)$, $x \in [2,3]$, is
- 1) $-\frac{3}{2} \log_{10} 3$ 2) $1 + \log_{10} 3$ 3) $\log_{10} 3$ 4) $\frac{3}{2} \log_{10} 3$
71. Assertion (A) : The angle between the curves $y = \sin x$, $y = \cos x$ is $\tan^{-1} 2\sqrt{2}$.
- Reason (R) : The slopes of their tangents at $\left(\frac{\pi}{4}, \frac{1}{\sqrt{2}}\right)$ are $\frac{1}{\sqrt{2}}$ and $-\frac{1}{\sqrt{2}}$ respectively.
- 1) Both A and R are true R is the correct explanation of A
 2) Both A and R are true R is not the correct explanation of A
 3) A is true but R is false
 4) A is false but R is true

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72. If $3x + 2y = 1$ is a tangent to $y = f(x)$ at $x = 1/2$, then $\lim_{x \rightarrow 0} \frac{x(x-1)}{f\left(\frac{e^{2x}}{2}\right) - f\left(\frac{e^{-2x}}{2}\right)}$
- 1) $1/3$ 2) $1/2$ 3) $1/6$ 4) $1/7$
73. The number of points on the curve $y = 54x^5 - 135x^4 - 70x^3 + 180x^2 + 210x$ at which the normal lines are parallel to $x + 90y + 2 = 0$.
- 1) 0 2) 1 3) 2 4) 4
74. If Roll's theorem holds for the function $f(x) = x^3 - ax^2 + bx - 4$, $x \in [1, 2]$ with $f'\left(\frac{4}{3}\right) = 0$, then ordered pair $(a, 2)$ is equal to:
- 1) $(5, 8)$ 2) $(-4, 8)$ 3) $(4, -8)$ 4) $(-5, -8)$
75. **Statement I :** If Rolle's theorem be applied in $f(x)$ then Lagrange's Mean Value theorem(LMVT) is also applied in $f(x)$
Statement II : Both Rolle's theorem and LMVT cannot be applied in $f(x) = |\sin |x||$ in $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$
- 1) Statement-I and Statement-II are true Statement-II is the correct explanation of Statement-I
 2) Statement-I and Statement-II are true Statement-II is not the correct explanation of Statement-I
 3) Statement-I is true but Statement-II is false
 4) Statement-I is false but Statement-II is true
76. A variable triangle is inscribed in a circle of radius R. if the rate of change of a side is $\sqrt{3}R$ times the rate of change of the opposite angle, then the opposite angle is
- 1) $\frac{\pi}{2}$ 2) $\frac{\pi}{6}$ 3) $\frac{\pi}{3}$ 4) $\frac{\pi}{4}$



77. **Assertion (A) :** The ordinate of a point describing the circle $x^2 + y^2 = 20$ decreases at the rate of 1.5 cm/s. The rate of change of the abscissa of the point in first quadrant when ordinate equals 4 cm is 3 cm/s.

Reason (R) : If $x^2 + y^2 = r^2$ then $x \frac{dx}{dt} + y \frac{dy}{dt} = 0$

- 1) Both A and R are true R is the correct explanation of A
- 2) Both A and R are true R is not the correct explanation of A
- 3) A is true but R is false
- 4) A is false but R is true

78. The function $f(x) = \frac{4x^3 - 3x^2}{6} - 2 \sin x + (2x - 1) \cos x$:

- 1) Increases in $\left[\frac{1}{2}, \infty\right)$
- 2) Increases in $\left(-\infty, \frac{1}{2}\right]$
- 3) decreases in $\left[\frac{1}{2}, \infty\right)$
- 4) decreases in $\left(-\infty, \frac{1}{2}\right]$

79. Let $f(x) = \tan^{-1}\left(\frac{1-x}{1+x}\right)$

Statement – I : The difference between the greatest and smallest value of $f(x)$ on $[0,1]$ is $\frac{\pi}{4}$.

Statement – II : If a function g decreases on $[a,b]$ then the greatest value of $g = g(a)$ and least value of $g = g(b)$.

- 1) Statement-I and Statement-II are true Statement-II is the correct explanation of Statement-I
- 2) Statement-I and Statement-II are true Statement-II is not the correct explanation of Statement-I
- 3) Statement-I is true but Statement-II is false
- 4) Statement-I is false but Statement-II is true

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80. $f(x) = (x-8)^4(x-9)^5, 0 \leq x \leq 10$, monotonically decreases in

- 1) $\left(\frac{76}{9}, 10\right]$ 2) $\left(8, \frac{76}{9}\right)$ 3) $[0, 8)$ 4) $\left(\frac{76}{9}, 10\right]$

(NUMERICAL VALUE TYPE)

Section-II contains 10 Numerical Value Type questions. Attempt any 5 questions only. First 5 attempted questions will be considered if more than 5 questions attempted. The Answer should be within 0 to 9999. If the Answer is in **Decimal** then round off to the **nearest Integer** value (Example i.e. If answer is above 10 and less than 10.5 round off is 10 and If answer is from 10.5 and less than 11 round off is 11).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

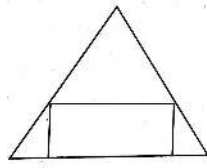
81. If the angle made by the tangent at the point (x_0, y_0) on the curve

$x = 12(t + \sin t \cos t), y = 12(1 + \sin t)^2, \left(0 < t < \frac{\pi}{2}\right)$ with the positive x-axis is $\frac{\pi}{3}$, then y_0 is equal to

82. If the curves $x = y^4$ and $xy = k$ cut at right angles, then $2 + (4k)^6$ is equal to _____.

83. If the tangent to the curve $y = x^3 - x^2 + x$ at the point (a, b) is also tangent to the curve $y = 5x^2 + 2x - 25$ at the point $(2, -1)$, then $3a + 9b$ is equal to _____.

84. If a rectangle is inscribed in an equilateral triangle of side length $2\sqrt{2}$ as shown in the figure, then the square of the largest area of such a rectangle is _____.



85. If the tangent to the curve, $y = e^x$ at a point (c, e^c) and the normal to the parabola. $y^2 = 4x$ at the point $(1, 2)$ intersect at the same point on the x-axis, then the value of c is _____.

86. The length of the sub-tangent to the hyperbola $x^2 - 4y^2 = 4$ at a point where slope of the normal is unity is $\frac{1}{\sqrt{k}}$, then k is equal to _____.

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87. A curve is defined parametrically by the equation $x = t^2$ and $y = t^3$. A variable pair of perpendicular lines through the origin O meet the curve at P and Q. If the locus of the point of intersection of the tangents at P and Q is $ay^2 = bx - 1$, then the value of $(a^2 + b^2)$ is _____
88. Consider $p(x)$ to be a polynomial of degree 5 having extremum at $x = -1, 1$ and $\lim_{x \rightarrow 0} \left(\frac{p(x)}{x^3} - 2 \right) = 4$. Then the value of $[P(2)]$ is (where $[.]$ represents greatest integer function)
89. If water is poured into an inverted hollow cone whose semi-vertical angle is 30° , its depth (measured along the axis) increases at the rate of 1 cm/s. Find the rate at which the volume of water increases when the depth is 24 cm is $k\pi \text{ cm}^3/\text{sec}$ then $k =$
90. A given right circular cone has a volume p , and the largest right circular cylinder that can be inscribed in the cone has a volume q . Then $\frac{8p}{9q} =$

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